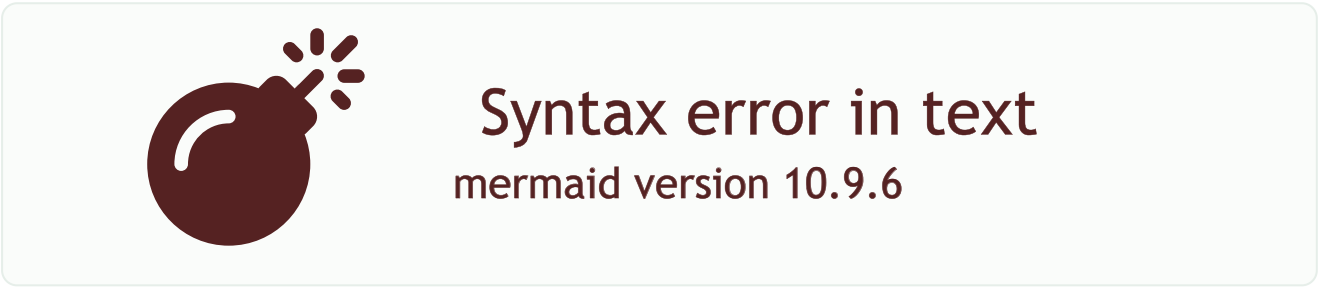
Smart AgriTech EMS Documentation

Backend reference

Node.js API for Smart AgriTech EMS: `ems/ems-backend/`.

Request pipeline



Mount order (`server.js`):

1. `GET /health` , `GET /metrics`
2. `POST /api/ingest/*` (no JWT)
3. `app.use('/api', apiLimiter, routes)`

Authentication & authorization

JWT auth (`middleware/auth.js`)

Header	Format
`Authorization`	`Bearer <accessToken>`

Role	Enum	Typical guards
Super Admin	`SUPER_ADMIN`	`authorize('SUPER_ADMIN')`
Org Admin	`ORG_ADMIN`	`authorize('SUPER_ADMIN', 'ORG_ADMIN')`
User	`USER`	Device-scoped in controllers

Ingest auth (`utils/ingestAuth.js`)

Header	Value
`x-api-key`	Global `INGEST_API_KEY` **or** per-device key (SHA-256 hash in DB)

Per-device keys returned **once** on `POST /api/devices` and `POST /devices/:id/regenerate-ingest-key` .

API route map

Auth — `/api/auth`

Method	Path	Auth
POST	`/login`	Public (rate limited)
POST	`/refresh`	Public
POST	`/logout`	Public
GET	`/me`	JWT
POST	`/change-password`	JWT
POST	`/forgot-password`	Public
POST	`/reset-password`	Public

Ingest — `/api/ingest`

Method	Path	Body
POST	`/`	`{ deviceId, slaveId?, readings[] }`
POST	`/command-ack`	`{ deviceId, commandId, status?, reason? }`

Devices — `/api/devices`

Method	Path	Roles (mutations)
GET/POST	`/`	POST: SUPER_ADMIN, ORG_ADMIN
GET/PUT/DELETE	`/:id`	PUT/DELETE: SUPER_ADMIN, ORG_ADMIN
PATCH	`/:id/switch`	SUPER_ADMIN, ORG_ADMIN

Method	Path	Roles (mutations)
POST	`/:id/regenerate-ingest-key`	SUPER_ADMIN, ORG_ADMIN
GET	`/:id/commands/:commandId`	Authenticated

Device config — `/api/devices/:deviceId/config`:

- `GET /slaves`, `GET /slaves/:id/variables`
- `PATCH /variables/:id`, `GET /variables/:id/log`

Device users — `/api/devices/:deviceId/users`:

- `GET`, `POST`, `DELETE /:userId`

Gateways — `/api/gateways`

Full CRUD; mutations: SUPER_ADMIN, ORG_ADMIN.

Device templates — `/api/device-templates`

- CRUD + `POST /:id/clone`
- Nested: `/slaves`, `/slaves/:slaveId/variables` CRUD

Sensor data — `/api/sensor-data`

Method	Path	Purpose
GET	`/latest`	Current values per variable (`?deviceId&slaveId`)
GET	`/history`	Single variable time series (`variableName` required)
GET	`/readings`	Paginated raw ingest rows
GET	`/aggregate`	Bucketed averages (`variableName` , `timeRange`)
GET	`/dashboard-summary`	KPIs + chart data for dashboards
GET	`/download`	CSV export
DELETE	`/`	SUPER_ADMIN bulk delete

Time ranges: `24h`, `7d`, `30d` (see `utils/helpers.js` `TIME_RANGE_MS`).

AI analytics — `/api/ai`

Path	Purpose
`/energy-consumption`	Consumption charts

Path	Purpose
`/power-factor`	PF trend + alarms
`/voltage-imbalance`	Phase voltage analytics
`/current-imbalance`	Current analytics
`/predictions`	Forecast-style readings

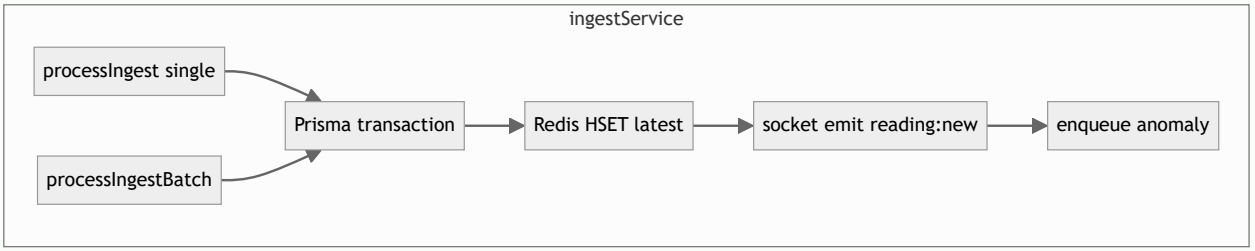
Alarms — `/api/alarm-*`, `/api/alarm-history`

- Templates, settings, contacts: CRUD
- History: variable alarms, linkage records, notifications
- `PATCH /alarm-history/variable-alarms/:id/process`

Other groups

Prefix	Features
`/api/organizations`	CRUD; `/me` for org admin
`/api/users`	CRUD, status, reset password
`/api/notifications`	List, read, delete
`/api/scheduled-tasks`	CRUD, toggle, logs
`/api/anomalies`	List, timeline, acknowledge
`/api/interval-history`	Billing intervals
`/api/slab-rates`	Tariff slabs
`/api/widget-templates`	Dashboard widgets
`/api/icons`, `/products`, `/themes`, `/settings`	Platform assets
`/api/subscriptions`	Plan requests
`/api/device-timestamps`	Connectivity log

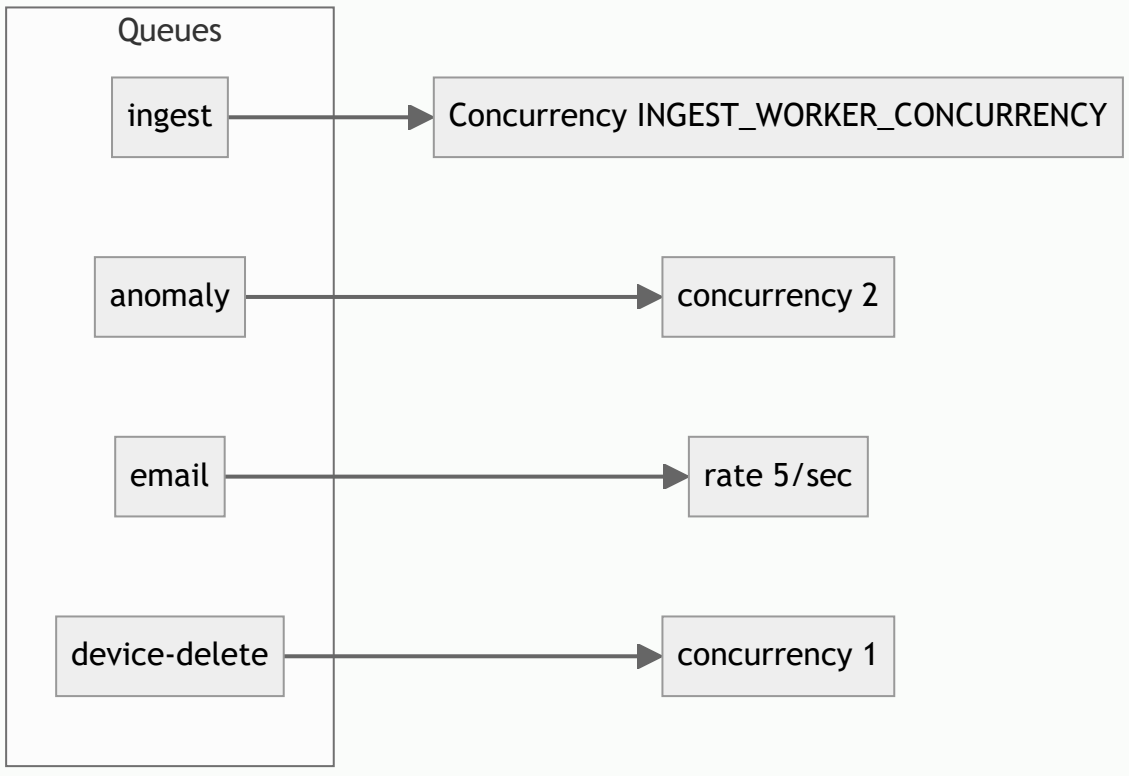
Ingest service (deep dive)



Writes per ingest:

1. `SensorReading` — JSON blob of all variables in batch
2. `SensorReadingValue` — normalized float rows (analytics indexes)
3. `Device.lastDataReceivedAt`, `DeviceTimestamp`
4. `DeviceConfigVariable.currentValue` (unless `SKIP_PG_CURRENT_VALUE=true`)
5. Redis `device:{id}:latest` hash

Workers (`workers/jobQueues.js`)



Queue	Trigger	Handler
`ingest`	HTTP ingest when Redis up	`processIngestBatch`
`anomaly`	After ingest	`runAnomalyCheck`
`email`	Alarm notification	Nodemailer send
`device-delete`	Device removal	Cascade purge

Socket.IO (`socket/index.js`)

Server → client events

Event	Room	Payload
`reading:new`	`device_{deviceId}`	`{ deviceId, ... }`
`device:command`	`device_{deviceId}`	Command status
`device:switch`	`org_{organizationId}`	Scheduler switch
`alarm:new`	`org_{organizationId}`	Alarm summary

Client → server

Event	Action
`join:device`	Subscribe to device room

Auth: `handshake.auth.token` must be valid JWT.

Database model summary

Domain	Models
Tenancy	`Organization` , `User` , `RefreshToken` , `Subscription`
IoT	`Gateway` , `Device` , `DeviceTemplate*` , `DeviceConfig*` , `DeviceUser` , `MqttConfig`
Telemetry	`SensorReading` , `SensorReadingValue` , `AIForecastReading`
Commands	`DeviceCommand`
Alarms	`TemplateTrigger` , `AlarmSetting` , `AlarmContact` , `DeviceVariableAlarmHistory` , `DeviceVariableLinkageHistory`

Domain	Models
Ops	`ScheduledTask`, `ScheduleExecutionLog`, `Notification`
Billing	`SlabRate`, `IntervalHistory`
UI	`WidgetTemplate`, `Icon`, `Theme`, `Product`, `SystemSetting`

Schema: `prisma/schema.prisma`
Migrations: `prisma/migrations/`

Environment variables

Variable	Required	Description
`DATABASE_URL`	Yes	PostgreSQL connection
`JWT_SECRET`	Yes	Token signing
`CLIENT_URL`	Yes prod	CORS origins (comma-separated)
`REDIS_URL`	Prod recommended	Queued ingest + cache
`INGEST_API_KEY`	Yes	Global ingest fallback
`PORT`	No	Default 5000 (CapRover: 9001)
`SKIP_PG_CURRENT_VALUE`	No	Default `true` with Redis
`INGEST_BATCH_MAX` / `INGEST_BATCH_MS`	No	Batch tuning
`INGEST_DEVICE_MAX_PER_MIN`	No	Per-device rate limit
`VALUE_FLUSH_MS`	No	Redis→PG flush interval
`DATABASE_READ_URL`	No	Read replica
`CLOUDINARY_*`	No	Image uploads
`SMTP_*` / `NODEMAILER_*`	No	Email
`PRISMA_LOG_QUERIES`	No	Debug only

Full list: `ems/ems-backend/.env.example`

Scripts & tooling

npm script	Purpose
`npm run dev`	Nodemon server
`npm run seed`	Demo users & sample data
`npm run seed:fleet`	100 devices / 10 gateways
`npm run simulate:production`	1s interval fleet simulator
`npx prisma migrate deploy`	Apply migrations

Docker

```
# ems/ems-backend/Dockerfile
# - node:20-alpine
# - prisma generate + migrate on entrypoint
# - EXPOSE 5000 (or CapRover PORT)
```

`docker-entrypoint.sh` runs `npx prisma migrate deploy` before `node server.js`.

Health & metrics

`GET /health`

```
{
  "status": "ok",
  "redis": true,
  "ingestMode": "queued"
}
```

`GET /metrics` — Prometheus text format (`utils/metrics.js`).

Related documents

- Architecture
- System flows
- Web frontend

- Deployment